

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	12
Experiment Name	Random Sampling and Sampling Distribution

DESCRIPTION

To perform the experiment: Random Sampling and Sampling Distribution.

CODE

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
population = np.random.normal(loc=50, scale=10, size=10000)
sample_size, repetitions = 30, 500
sample_means = []
for _ in range(repetitions):
    sample = np.random.choice(population, sample_size, replace=False)
    sample_means.append(sample.mean())
print("Population mean:", round(population.mean(), 2))
print("Mean of sample means:", round(np.mean(sample_means), 2))
plt.hist(sample_means, bins=25, edgecolor="black")
plt.title("Sampling Distribution of Sample Means")
plt.xlabel("Sample mean"); plt.ylabel("Frequency"); plt.show()
```

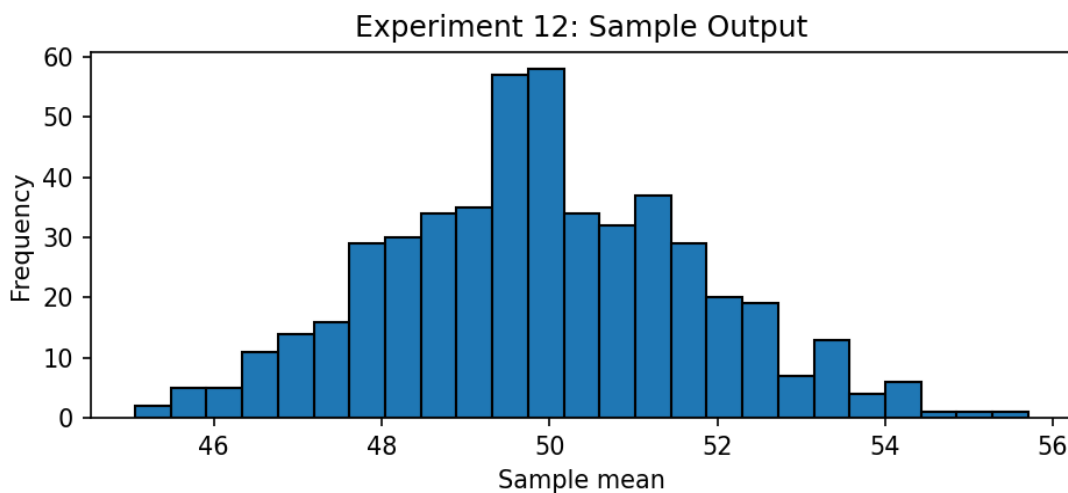
SAMPLE INPUT

Sample input: simulated population with mean 50 and standard deviation 10; sample size 30; 500 repetitions.

OUTPUT — SAMPLE CONSOLE OUTPUT

Population mean: approximately 50\nMean of sample means: approximately 50\nSampling distribution histogram displayed.

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature